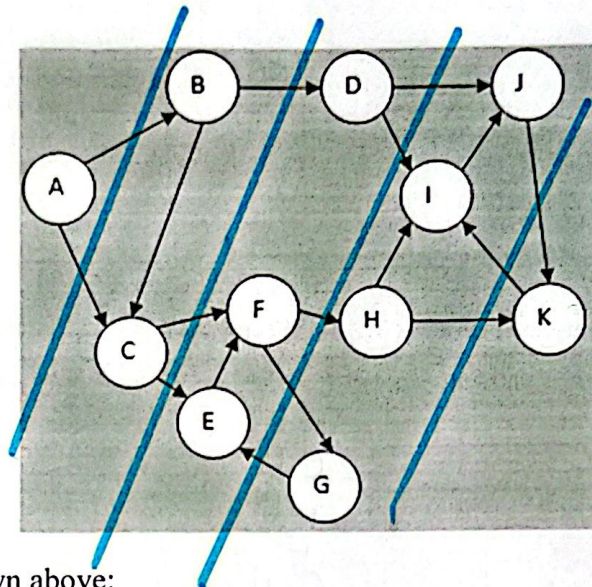




Consider the graph shown above:

Start Node: A

Goal Node: K

Cost per edge: 1

Expand children in left-to-right order.

Tasks

1. Perform **Breadth-First Search (BFS)** starting from A until the goal K is reached.

- Record the **traversed path (visit order)**
- Record the **cost**, calculated as:

Number of edges traversed until K is reached

2. Perform **Depth-First Search (DFS)** starting from A until the goal K is reached.

- Record the **traversed path (visit order)**
- Record the **cost**

Fill the following table:

BFS Table

| Step | Node Expanded | Children Generated | Frontier (Queue) |
|------|---------------|--------------------|------------------|
| 1    |               |                    | A                |
| 2    | A             | CB                 | CB               |
| 3    | C             | EF                 | BEF              |
| 4    | B             | CD                 | EFD              |
| 5    | E             | F                  | FD               |
| 6    | F             | GH                 | DGH              |
| 7    | D             | IJ                 | GHIJ             |
| 8    | G             | E                  | HIT              |
| 9    | H             | IK                 | ITK              |
| 10   |               |                    |                  |

Traversal Path: ACBEFDGHIJK

Traversal Cost: 10

Goal Check

Goal

DFS Table

Goal  
if check

| Step | Node Expanded | Children Generated                             | Stack |
|------|---------------|------------------------------------------------|-------|
| 1    |               |                                                | A     |
| 2    | A             | CB                                             | CB    |
| 3    | C             | EF                                             | EFB   |
| 4    | E             | F                                              | FB    |
| 5    | F             | GH                                             | GH B  |
| 6    | G             | E                                              | H B   |
| 7    | H             | KI                                             | KI B  |
| 8    | K             |                                                |       |
| 9    |               | Traversal path : ACEFGHK<br>Traversal cost : 6 |       |
| 10   |               |                                                |       |

Goal.

| Algorithm | Traversed Path (Visit Order) | Cost |
|-----------|------------------------------|------|
| BFS       | ACBEFDGHIJK                  | 10   |
| DFS       | ACEFGHK                      | 6    |